



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

- Summary of methodologies
 - Data Collection with APIs
 - Data Collection with Webscraping
 - Data Wrangling
 - Exploratory Data Analysis with SQL
 - EDA with Data Visualization
 - Interactive Visual Analytics with Folium and Dashboard
 - Predictive Analytics and Machine Learning
- Summary of all results
 - EDA Results
 - Interactive Analytics
 - Predictive Analytics Results

Introduction

- Project background and context
 - The evolution and placement that SpaceX has in the space travel and exploration industries has been a huge part of how the company has been so boundary-pushing. Being able to move into the next step of the evolution for SpaceX's outer-exploration endeavors is the cost-savings of the first stage launch for the Falcon 9 rocket. If we can successfully predict the first stage launch for Rocket 9, it will help the business better understand and predict costs associated with the project.
- Problems you want to find answers
 - What variables are best correlated to predict a successful rocket launch?
 - Which best-selected variables show the best conditions and best interactions from the data provided?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - The data for this project was collected from SpaceX via their API and scraping the web from Wikipedia online.
- Perform data wrangling
 - One-hot encoding
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models

Data Collection

- Describe how data sets were collected.
 - The data for this project was collected via the request to the SpaceX API and web scraping from Wikipedia.
 - Data was cleaned by handling the missing values where there was missing fields in the datasets.

Data Collection – SpaceX API

- The SpaceX API was used to request the data we collected and cleaned for the project.
- GitHub Notebook link: <https://github.com/miller656565/PublicRepo/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

Task 1: Request and parse the SpaceX launch data using the GET re
To make the requested JSON results more consistent, we will use the following stati

```
[12]: static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud  
We should see that the request was successfull with the 200 status response code  
[13]: response=requests.get(static_json_url)  
[14]: response.status_code  
[14]: 200
```

Task 2: Filter the dataframe to only include Falcon 9 launches

Finally we will remove the Falcon 1 launches keeping only the Falcon 9 launches. Filter
keep the Falcon 9 launches. Save the filtered data to a new dataframe called data_f

```
: # Hint data['BoosterVersion']!='Falcon 1'  
data_falcon9 = df.loc[df['BoosterVersion']!='Falcon 1']
```

Now that we have removed some values we should reset the FlightNumber column

```
: data_falcon9.loc[:, 'FlightNumber'] = list(range(1, data_falcon9.shape[0]+1))  
data_falcon9
```

Task 3: Dealing with Missing Values

Calculate below the mean for the PayloadMass using the .mean(). Then us
the data with the mean you calculated.

```
: # Calculate the mean value of PayloadMass column  
mean = data_falcon9['PayloadMass'].mean()  
# Replace the np.nan values with its mean value  
data_falcon9['PayloadMass'].fillna(mean, inplace=True)  
data_falcon9.isnull().sum()
```


Data Collection - Scraping

- Using the GET method, we requested the Falcon9 launch information from Wikipedia to scrap the data.
- The data table from Wikipedia was parsed to retrieve the data and converted into a Pandas DataFrame.
- Link to GitHub notebook: <https://github.com/miller656565/PublicRepo/blob/main/jupyter-labs-webscraping.ipynb>

TASK 1: Request the Falcon9 Launch Wiki page from its URL

First, let's perform an HTTP GET method to request the Falcon9 Launch HTML page.

```
# use requests.get() method with the provided static_url
# assign the response to a object
data = requests.get(static_url).text
```

Create a `BeautifulSoup` object from the HTML `response`

```
# Use BeautifulSoup() to create a BeautifulSoup object from a response text content
soup = BeautifulSoup(data, 'html.parser')
```

Print the page title to verify if the `BeautifulSoup` object was created properly

```
# Use soup.title attribute
soup.title
```

```
<title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>
```

TASK 2: Extract all column/variable names from the HTML table header

Next, we want to collect all relevant column names from the HTML table header

Let's try to find all tables on the wiki page first. If you need to refresh your memory about `BeautifulSoup`

```
# Use the find_all function in the BeautifulSoup object, with element type 'table'
# Assign the result to a list called 'html_tables'
html_tables = soup.find_all('table')
```

Starting from the third table is our target table contains the actual launch records.

```
# Let's print the third table and check its content
first_launch_table = html_tables[2]
print(first_launch_table)
```

```
<table class="wikitable plainrowheaders collapsible" style="width: 100%;">
```

TASK 3: Create a data frame by parsing the launch HTML

We will create an empty dictionary with keys from the extracted column names

```
launch_dict = dict.fromkeys(column_names)

# Remove an irrelevant column
del launch_dict['Date and time ( )']

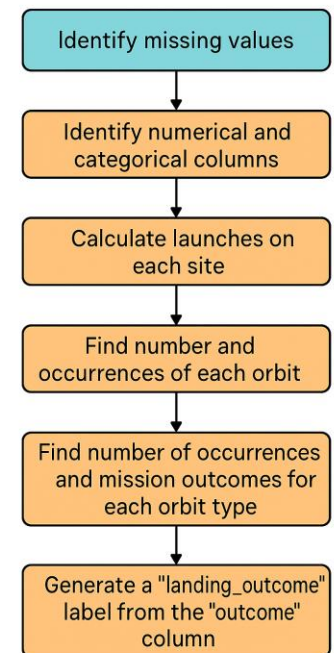
# Let's initial the launch_dict with each value to be an empty list
launch_dict['Flight No.']= []
launch_dict['Launch site']= []
launch_dict['Payload']= []
launch_dict['Payload mass']= []
launch_dict['Orbit']= []
launch_dict['Customer']= []
launch_dict['Launch outcome']= []

# Added some new columns
launch_dict['Version Booster']= []
launch_dict['Booster landing']= []
launch_dict['Date']= []
launch_dict['Time']= []
```

Data Wrangling

- Exploratory Data Analysis was performed on the data to determine where there were potential insights and patterns in the data to show which could be used to train the supervised models.
- GitHub Notebook link: <https://github.com/miller656565/PublicRepo/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

Data Wrangling EDA



EDA with Data Visualization

- Catplot: showed the relationship between the flight number and launch site
- Scatterplot: showed relationship between payload mass and launch site
- Barplot: showed relationship between success rate for each orbit type
- Catplot: showed relationship between flight number and orbit type
- Catplot: showed relationship between payload mass and orbit type
- Line chart: showed relationship between year and success rate
- Link to GitHub Notebook:
<https://github.com/miller656565/PublicRepo/blob/main/edadataviz.ipynb>

EDA with SQL

- Display the names of the unique launch sites in the space mission:
 - %sql SELECT distinct Launch_Site from spacetable
- Display 5 records where the launch sites begin with the string 'CCA':
 - %sql SELECT * from spacetable where launch_site like 'CCA%' limit 5
- Display the total payload mass carried by boosters launched by NASA (CRS):
 - %sql SELECT sum(payload_mass__kg_) from spacetable where customer = 'NASA (CRS)'
- Display average payload mass carried by booster version F9 v1.1:
 - %sql select avg(payload_mass__kg_) from spacetable where booster_version like 'F9 v1.1%'
- List the date when the first succesful landing outcome in ground pad was achieved:
 - %sql select min(date) from spacetable where landing_outcome = 'Success (ground pad)'
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
 - %sql select booster_version from spacetable where landing_outcome = 'Success (drone ship)' and payload_mass__kg_ >4000 and payload_mass__kg_ < 6000

EDA with SQL

- List the total number of successful and failure mission outcomes:
 - %sql select mission_outcome, count(*) from spacetable
- List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function:
 - %sql select BOOSTER_VERSION from SPACEXTBL where PAYLOAD_MASS__KG_ = (select max(PAYLOAD_MASS__KG_) from SPACEXTBL)
- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015:
 - %sql SELECT MONTH(DATE),MISSION_OUTCOME,BOOSTER_VERSION,LAUNCH_SITE FROM SPACEXTBL where EXTRACT(YEAR FROM DATE)='2015';
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order:
 - %sql select * from SPACEXTBL where Landing_Outcome like 'Success%' and (DATE between '2010-06-04' and '2017-03-20') order by date desc
- GitHub notebook link: https://github.com/miller656565/PublicRepo/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

- Map objects and Folium map summary
 - Folium circle and marker functions used to add the highlighted areas in the circle to label with text specific coordinates for every launch site
 - Simplification done by using the marker cluster method for the map coordinates
 - PolyLine method connects the closest objects (i.e. city or highway) between the launch sites to each other with a line drawn on the map
 - The mouse position method shows the coordinators for each mouse hover on the map
- GitHub notebook link:
https://github.com/miller656565/PublicRepo/blob/main/lab_jupyter_launch_site_location.ipynb

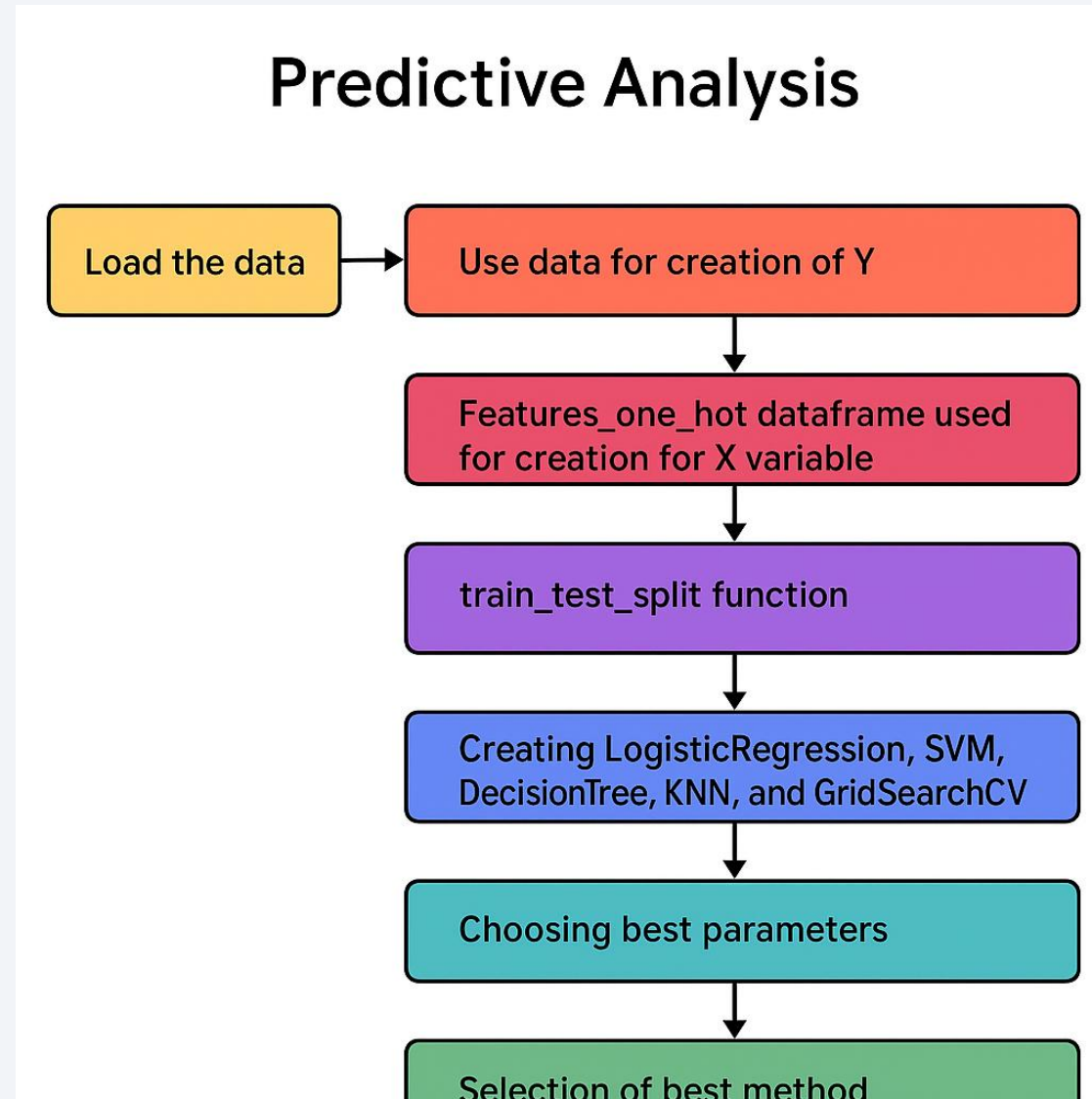
Build a Dashboard with Plotly Dash

- The dashboard app has components like a dropdown list, range slider for the pie chart interaction, and scatterplot chart.
 - Launch site dropdown filter
 - Pie charts to show success rate for launches
 - Scatterplot chart to show launch site, payload mass, and launch success
 - Range slider for selecting the payload mass
- GitHub notebook link:
<https://github.com/miller656565/PublicRepo/blob/main/spacex-dash-app.py>

Predictive Analysis (Classification)

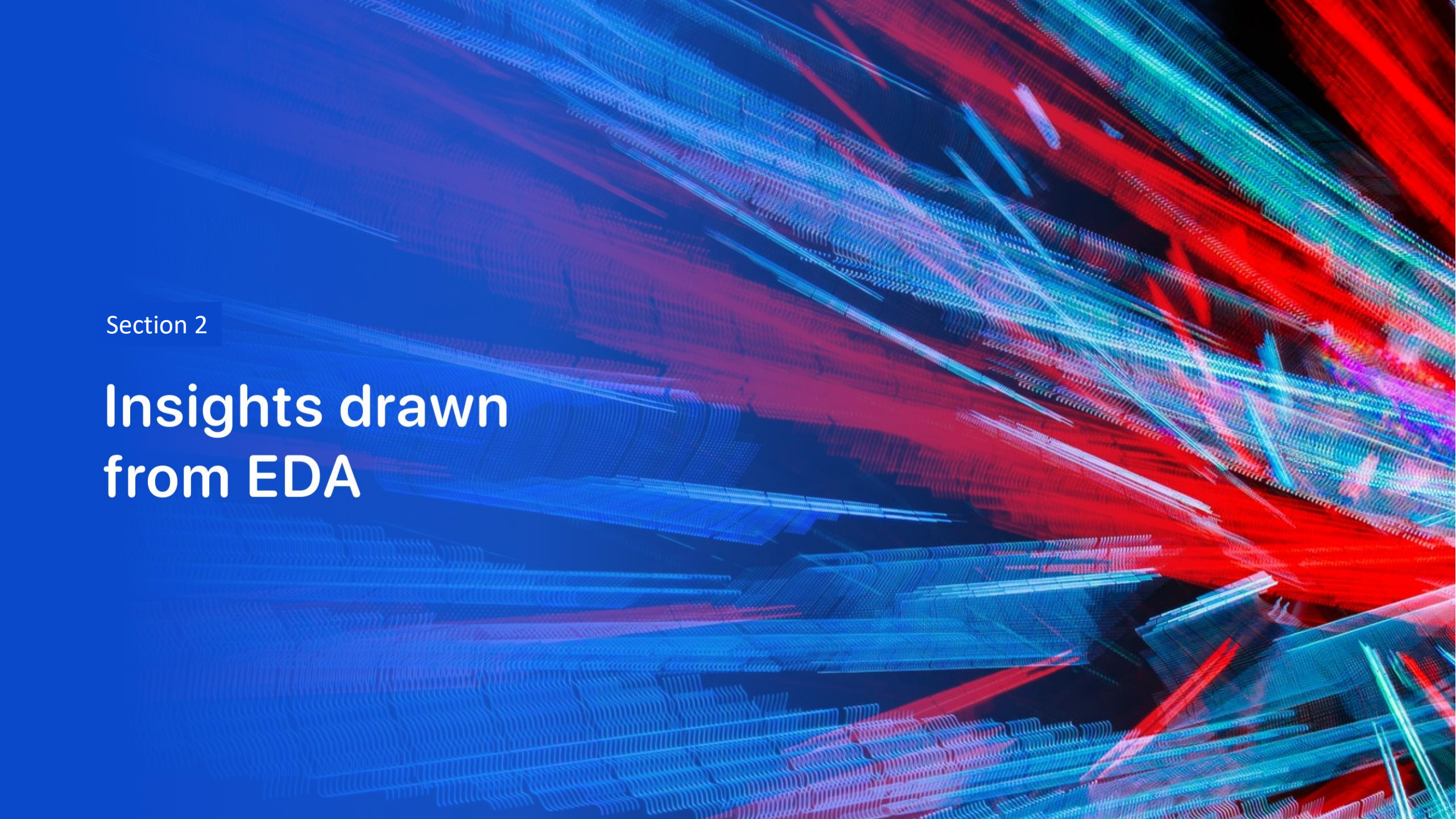
- This model was developed to predict if the first stage launch will land based on datasets from earlier in this lab.
 - NumPy array created to create class data
 - Standardization of the data
 - train_test_split function used to split training and test data
 - Tuning and search for best parameters for the model
 - Find model and method that best fits using the testing data
- GitHub notebook link:
https://github.com/miller656565/PublicRepo/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Predictive Analysis (Classification)



Results

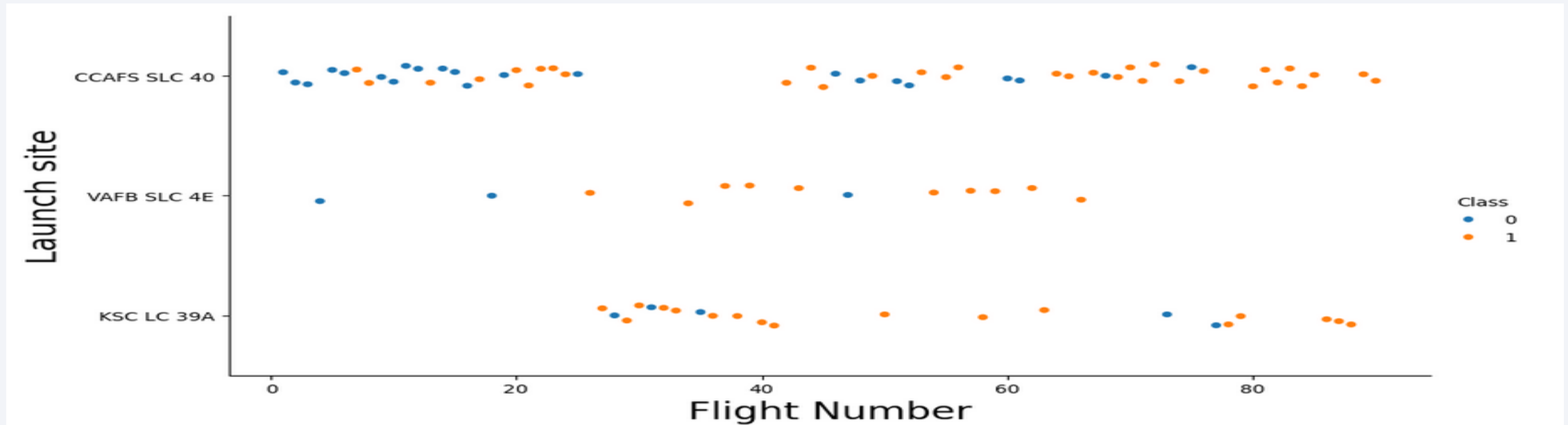
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the upper right quadrant. The overall effect is dynamic and technological.

Section 2

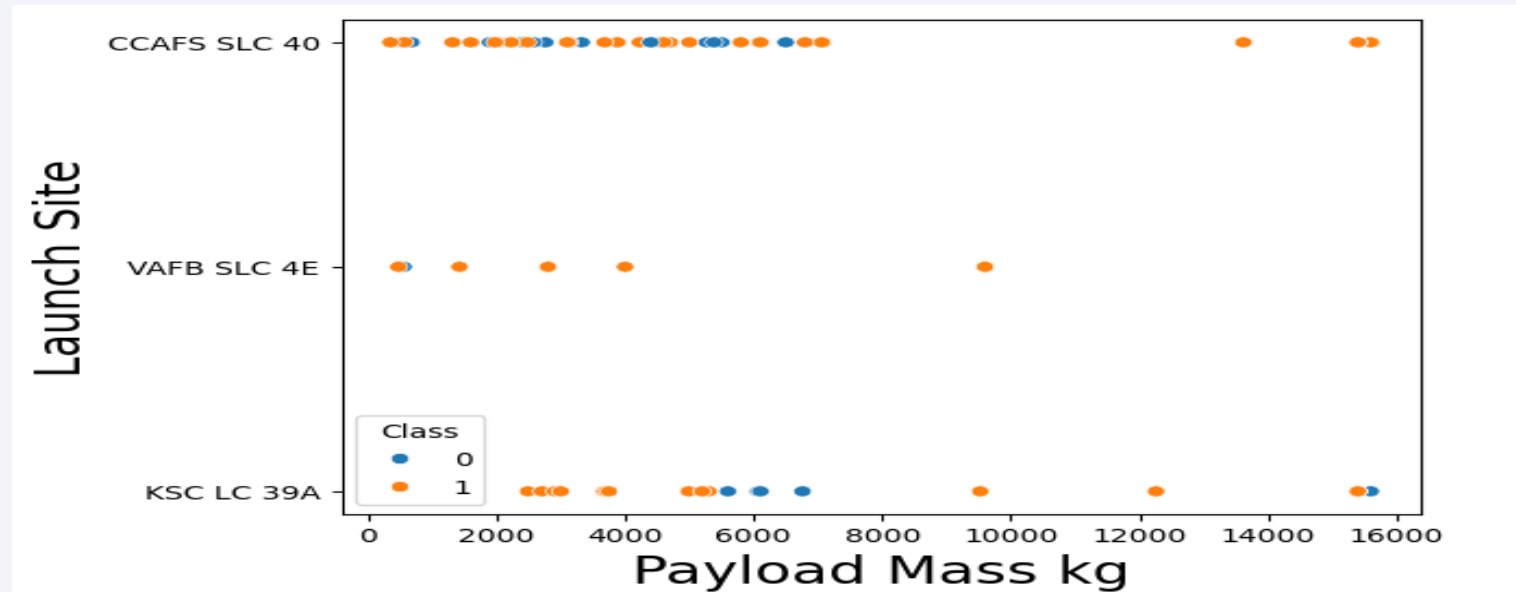
Insights drawn from EDA

Flight Number vs. Launch Site



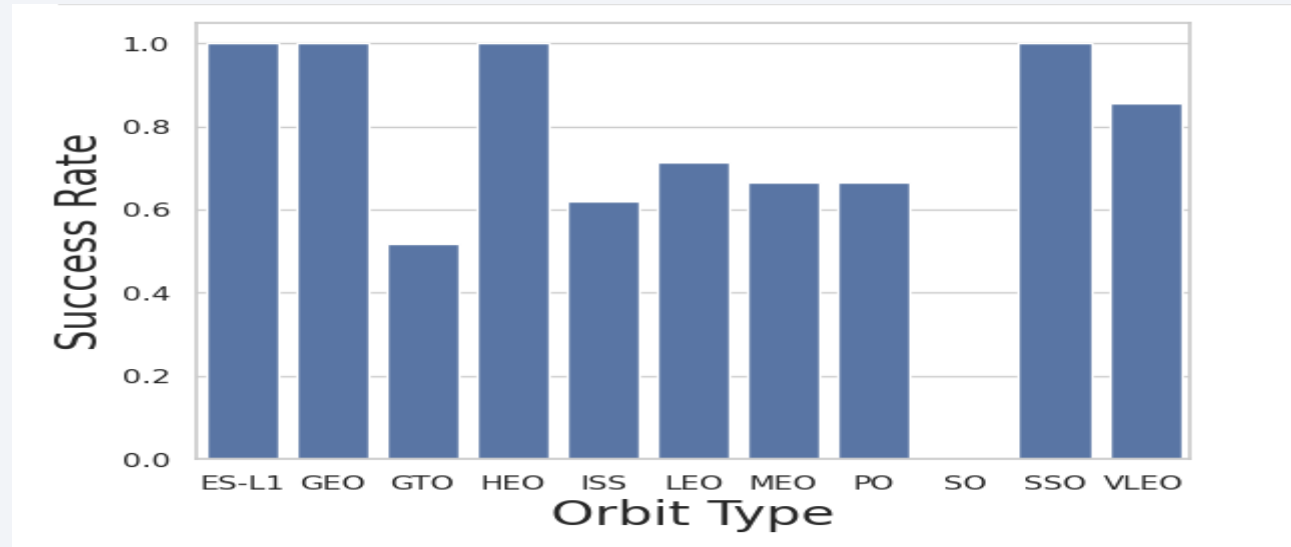
- The initial success rate of the launches was not very good for the three different launches. But it did improve over time.
- The best example of improving success rate over time is flight CCAFS SLC 40, which saw a much higher success rate after the first dozen launches, especially compared to launch sites VAFB SLC 4E and KSC LC 39A.

Payload vs. Launch Site



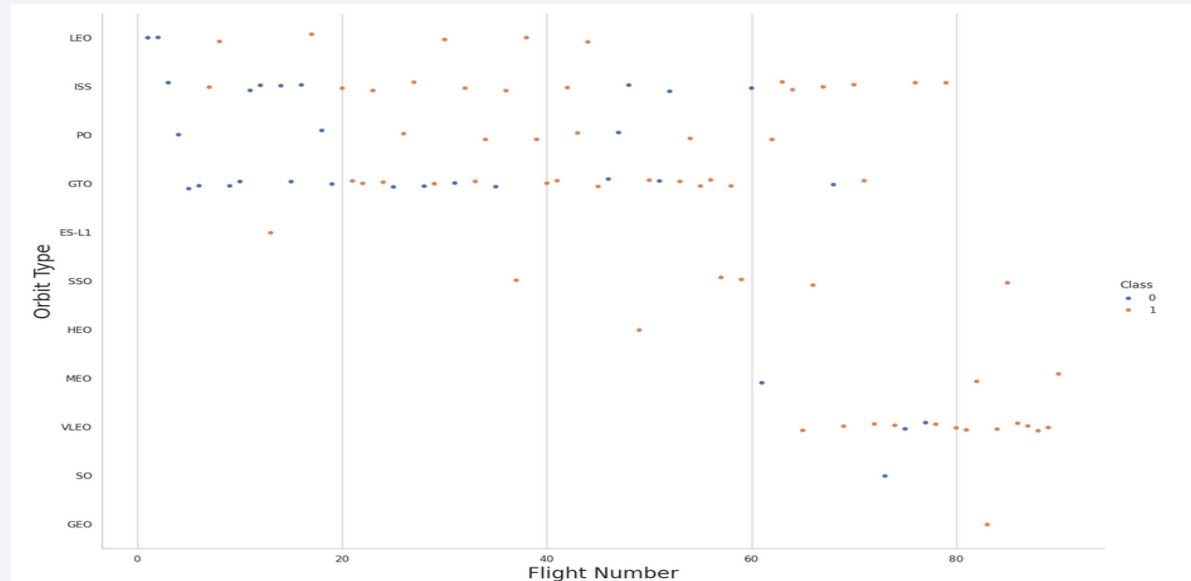
- Launch site VAFB SLC 4E didn't really have any heavy payload rockets. The heaviest payloads were limited at 10,000 KG.
- There wasn't a significant correlation between payload mass for kg and launch site successes.
- KSC LC 39A launch site didn't have any lower payload masses (under 2000 KG). Most of CCAFS SLC 40's launches were with low to medium payloads (under 8000 KG), but a handful were over 14000 KG.

Success Rate vs. Orbit Type



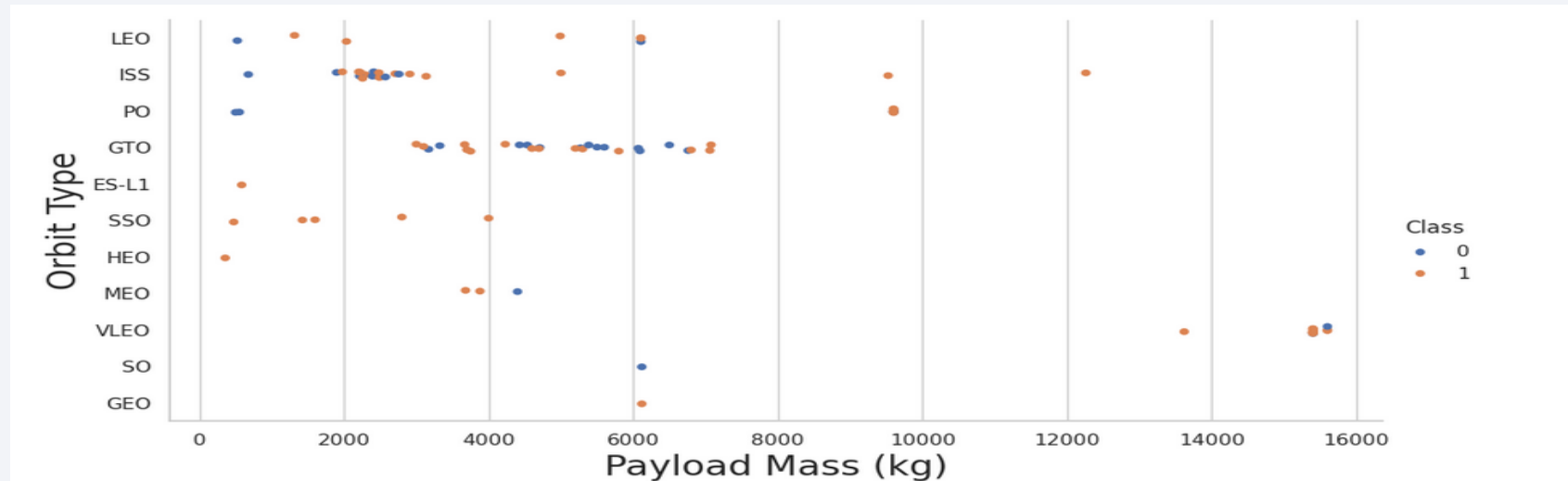
- ES-L1, SSO, HEO, and GEO had the best success rate among the different orbit types.
- Sample size of the datasets for the different orbit types could be impacting or skewing the data shown in the bar chart, especially for orbit type SO and GTO.

Flight Number vs. Orbit Type



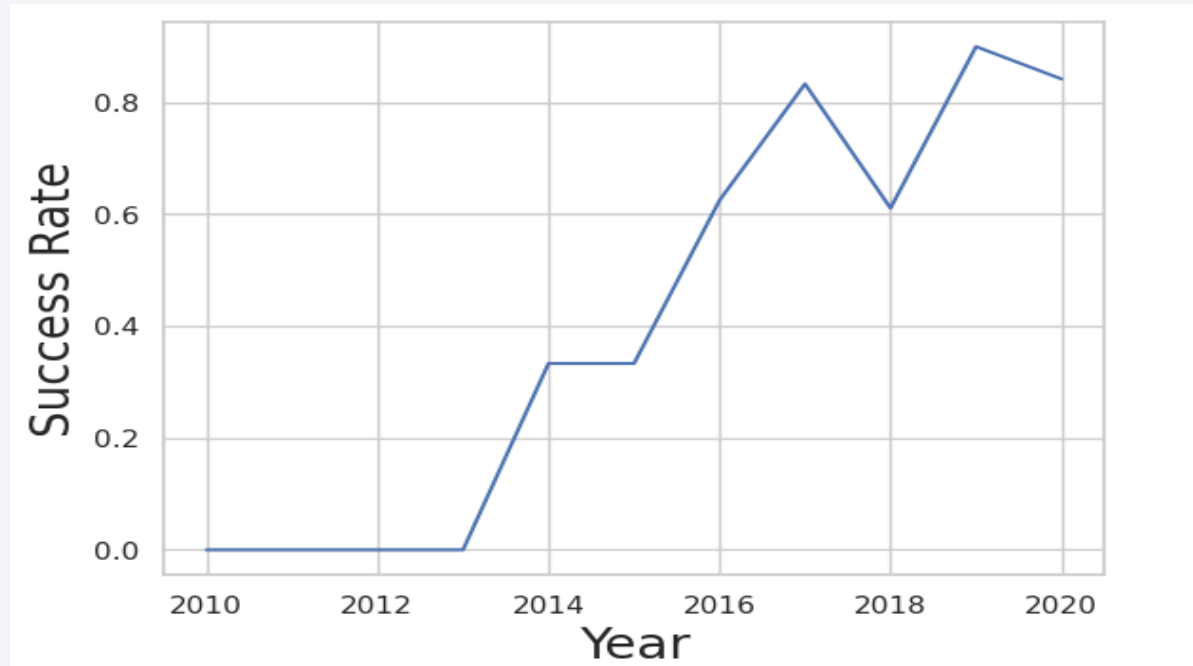
- Similar to the Flight Number vs. Launch Site plot, this chart shows that the success rate was higher for the launches over time.
- The orbit types with the highest success rate tend to have fewer total number of launches.

Payload vs. Orbit Type



- The launches with heavier payloads, especially for PO, ISS, and LEO, all have correlations with a more successful landing rate with the larger payloads.
- It's difficult to distinguish the success rate of landings for GTO and ISS in the more clustered outcomes.

Launch Success Yearly Trend



- The success rate trend was increasing from 2013 until 2020.

All Launch Site Names

```
: %sql SELECT distinct Launch_Site from spacetable
* sqlite:///my_data1.db
Done.
: Launch_Site
-----
    CCAFS LC-40
    VAFB SLC-4E
    KSC LC-39A
    CCAFS SLC-40
```

- There are four unique launch site names.
- I pulled the unique names from the launch site column in the table using the distinct function.

Launch Site Names Begin with 'CCA'

```
%sql SELECT * from spacetable where launch_site like 'CCA%' limit 5
```

```
* sqlite:///my_data1.db
```

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (paracl
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (paracl
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No att
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No att
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No att

- The limit and like functions were used with the where statement to pull in 5 records with launch sites beginning with 'CCA'.

Total Payload Mass

```
%sql SELECT sum(payload_mass_kg_) from spacetable where customer = 'NASA (CRS)'
```

```
* sqlite:///my_data1.db  
Done.
```

sum(payload_mass_kg_)
45596

- The sum function was used on the payload mass kg column, with a where statement filtering down the customer column, to find the total payload mass for 'NASA (CRS)' boosters launched from that customer.

Average Payload Mass by F9 v1.1

```
%sql select avg(payload_mass_kg_) from spacetable where booster_version like 'F9 v1.1'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

avg(payload_mass_kg_)

2534.6666666666665

- I was able to determine that the average payload mass for the booster version F9 v1.1 was 2534.67 kg by using the average and like functions with a where statement.

First Successful Ground Landing Date

```
%sql select min(date) from spacetable where landing_outcome = 'Success (ground pad)'
```

```
* sqlite:///my_data1.db  
Done.
```

min(date)
2015-12-22

- Using a min function on the date column and a where statement on the landing outcome field, I determined that the first successful landing outcome on a ground pad was 12-22-2015.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql select booster_version from spacetable where landing_outcome = 'Success (drone ship)' and payload_mass__kg_ >4000 and
* sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- The names of the boosters that had drone ships successful landings and payload masses < 6000 KG and > 4000 KG were found by filtering the landing outcome column in the where statement and the payload mass KG column with an and statement.

Total Number of Successful and Failure Mission Outcomes

```
%sql select mission_outcome, count(mission_outcome) as totaloutcomes from spacextbl group by mission_outcome
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Mission_Outcome	totaloutcomes
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- There were 99 successful, 1 failed, and 1 status unclear mission outcomes, which was found using the count and group by functions.

Boosters Carried Maximum Payload

```
%sql select Booster_Version from SPACEXTABLE where PAYLOAD_MASS_KG_ = (select max(PAYLOAD_MASS_KG_)from SPACEXTABLE)

* sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

- The names of the booster versions that have carried the max payload mass were found using a subquery with the max function.

2015 Launch Records

```
%sql SELECT date, substr(Date, 6,2), launch_site, booster_version, landing_outcome FROM spacextbl WHERE substr(Date,0,5)='2015' AND landing_outcome = 'Failure (drone ship)'
```

```
* sqlite:///my_data1.db
```

Done.

Date	substr(Date, 6,2)	Launch_Site	Booster_Version	Landing_Outcome
2015-01-10	01	CCAFS LC-40	F9 v1.1 B1012	Failure (drone ship)
2015-04-14	04	CCAFS LC-40	F9 v1.1 B1015	Failure (drone ship)

- I used the substr functions to narrow down month names and year filters in the where statement, along with an and statement to filter down the landing_outcome field to find all 2015 failed drone ship landings in 2015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql select Landing_Outcome, count(*) as 'Count' from SPACEXTABLE where Date between '2010-06-04' and '2017-03-20' group by Landing_Outcome order by Count desc
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

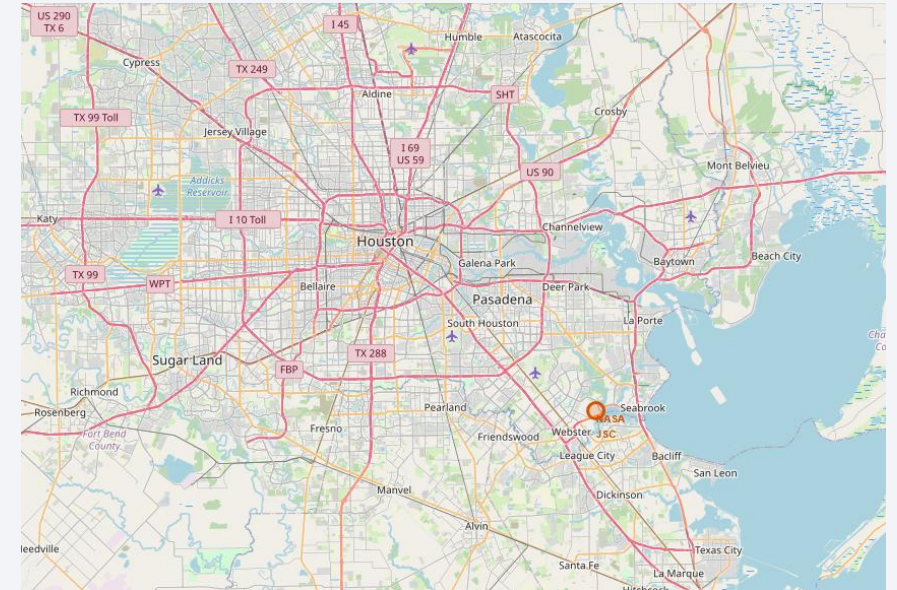
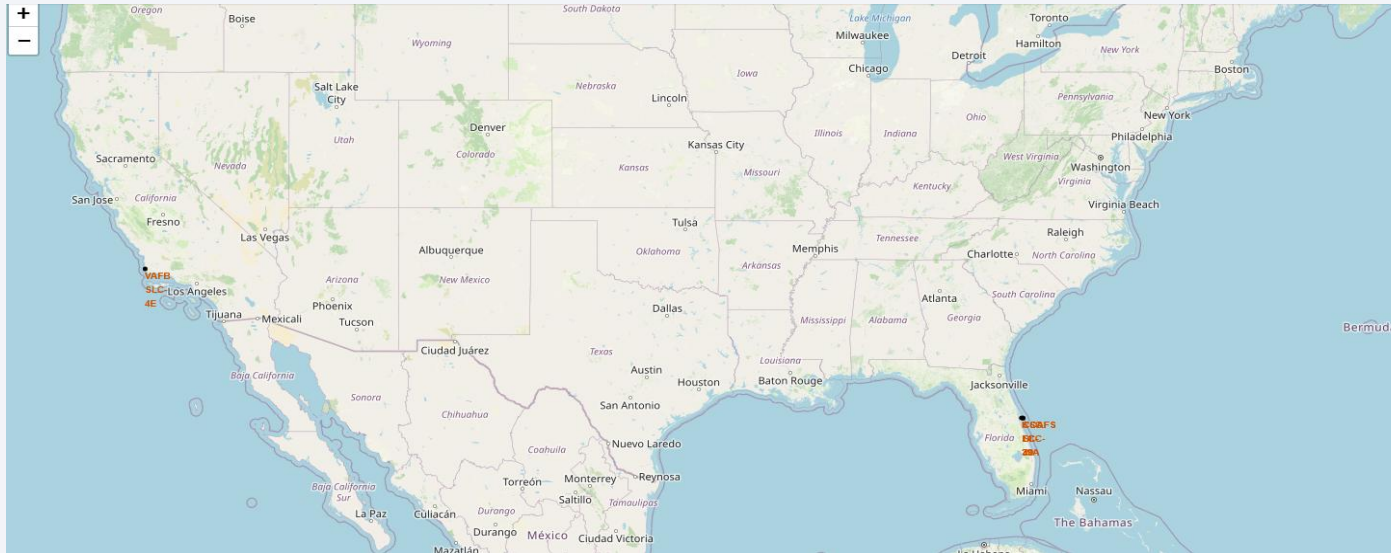
- The ranking and count of landing outcomes was found using the count function, along with (between and and) functions in the where statement, group by statement for the landing_outcome field, and then sorted by the order by statement.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

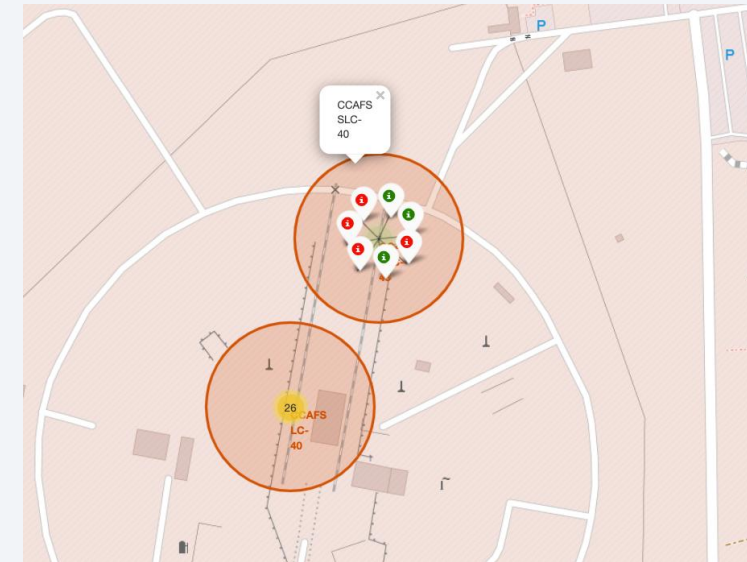
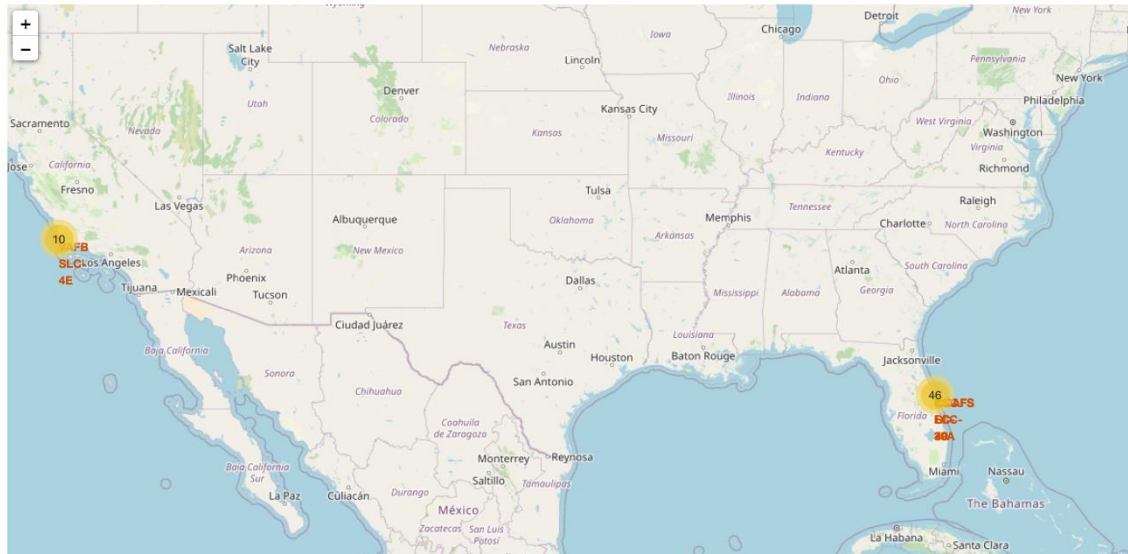
Launch Sites Proximities Analysis

Launch Sites



- The launch sites are, for the most part, close together in proximity on the east and west coasts.
- Explore the generated folium map and make a proper screenshot to include all launch sites' location markers on a global map
- Explain the important elements and findings on the screenshot

Success Rates for each Launch Site

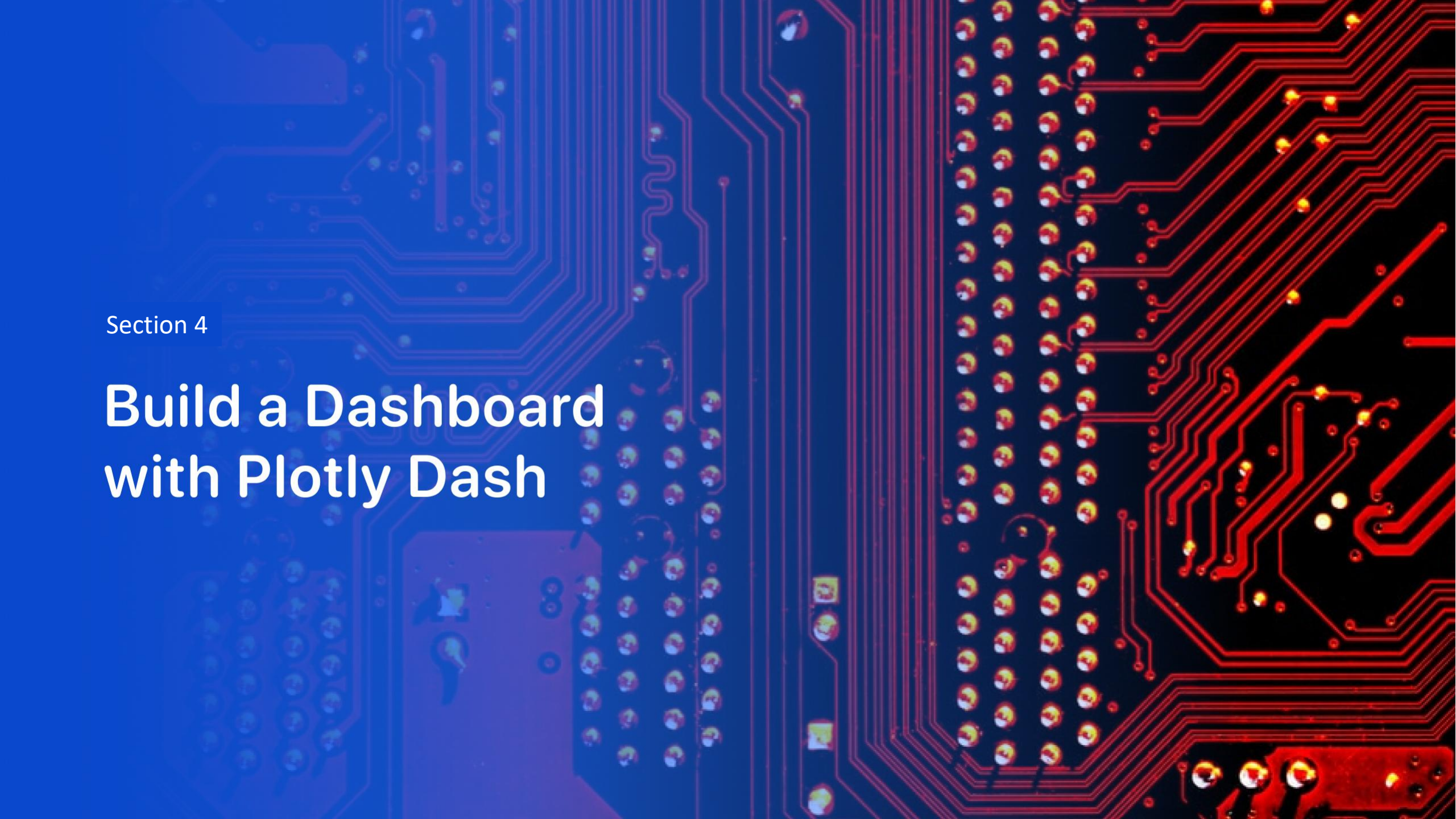


- The success and fail rates of the launches are shown in green and red markers. The areas of the launch sites are shown on the first map.

Proximities for each Launch Site



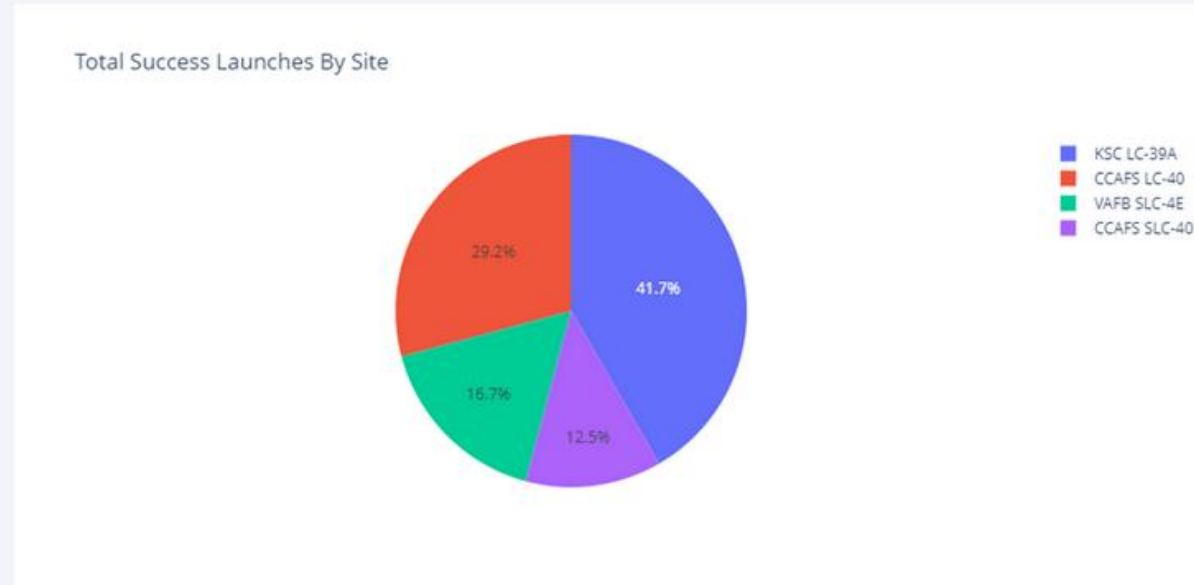
- The launch sites are often close to the coast, and located not too far from infrastructure access, including highways and train tracks.



Section 4

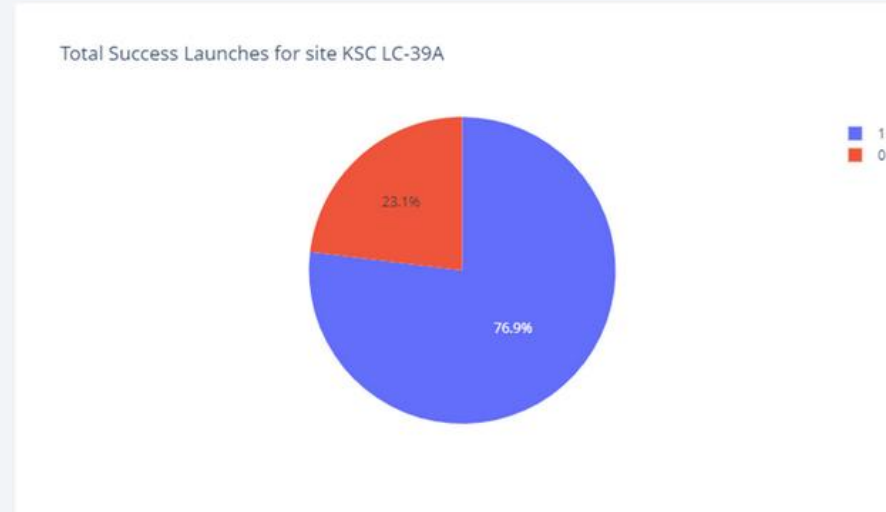
Build a Dashboard with Plotly Dash

Pie Chart for Successful Launches by Launch Site



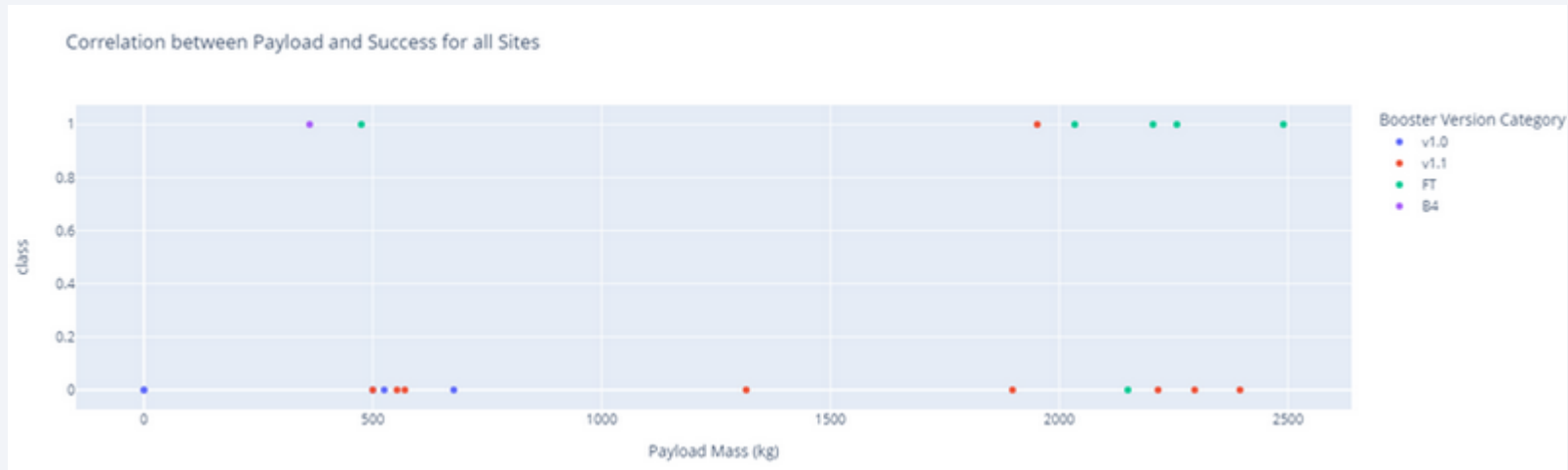
- The top two sites with the highest success rates for launches are KSC LC-39A and then CCAFS LC-40.

Pie Chart for KSC LC-39A



- This pie chart shows the ratio of success to failure for the launch site with the highest success rate.

Chart for Payload Mass vs. Launch Outcome



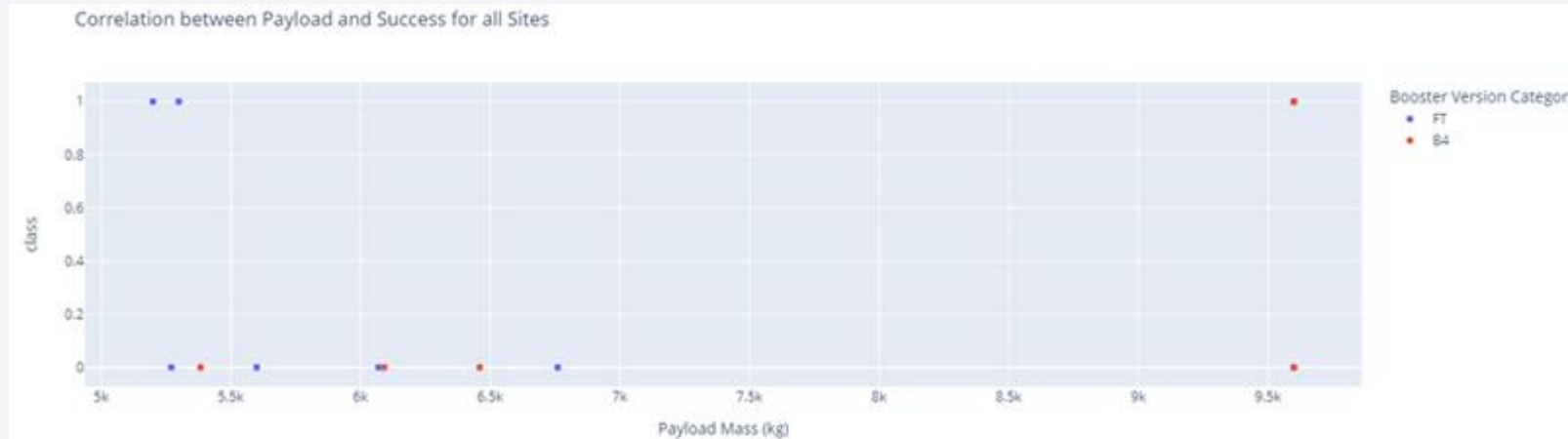
- The payload mass range of 2000-3000 KG has the most clustered outcome of successful launches on this chart.
- This is the chart for the 2500 KG range for launch outcomes.

Chart for Payload Mass vs. Launch Outcome



- The payload mass range of 4000-4500 KG has the most clustered outcome of failed launches on this chart.
- This is the chart for the 2500-5000 KG range for launch outcomes.

Chart for Payload Mass vs. Launch Outcome



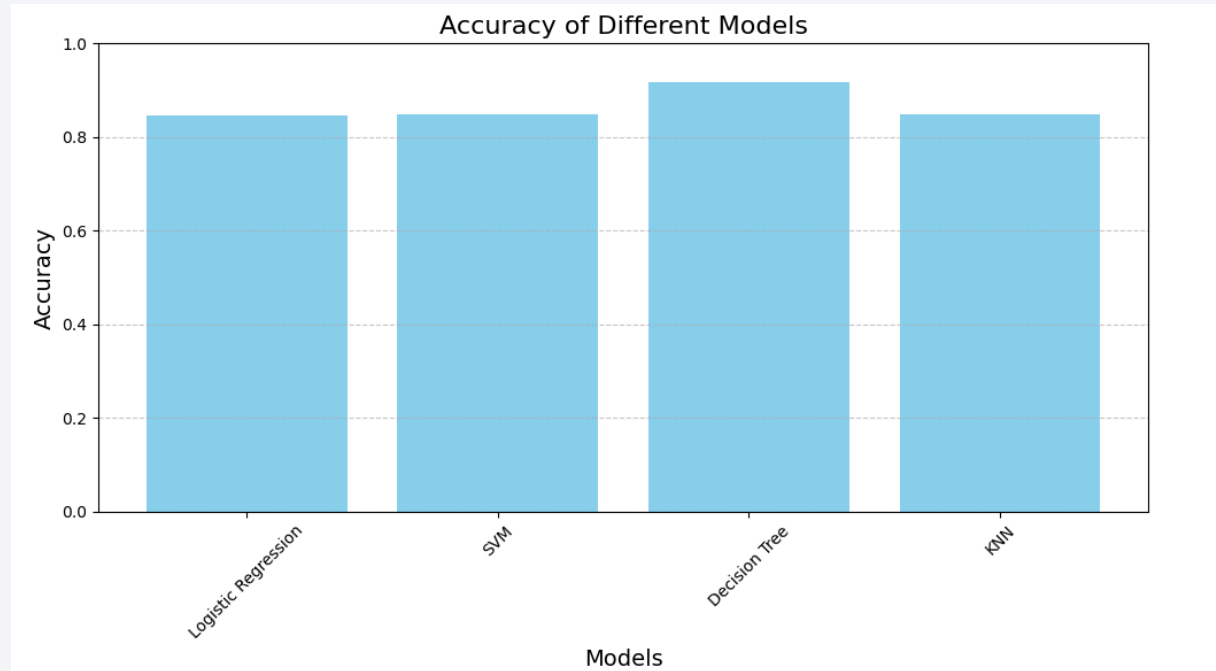
- Most of the launches in this range resulted in failures.
- This is the chart for the 5000-10000 KG range for launch outcomes.



Section 5

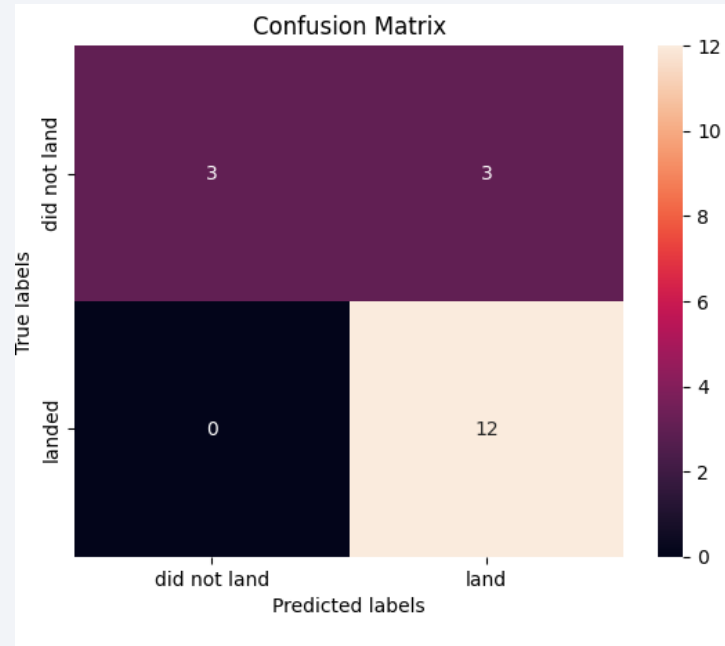
Predictive Analysis (Classification)

Classification Accuracy



- The Decision Tree was the most the best performing model, with an accuracy score of ~ 0.918 .

Confusion Matrix



- The Decision Tree model confusion matrix shows that this model predicted three true landings.

Conclusions

- Over time, the SpaceX launches performed better in terms of success rates for successful landings.
- There were more successful landings over time from 2013-2020.
- The KSC LC-39A launch site had the greatest success rate of successful landings.
- The Decision Tree was the best performing model.

Thank you!

